

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

04

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

In the xy -plane, a circle with radius 5 has center $(-8, 6)$. Which of the following is an equation of the circle?

A. $(x - 8)^2 + (y + 6)^2 = 25$

B. $(x + 8)^2 + (y - 6)^2 = 25$

C. $(x - 8)^2 + (y + 6)^2 = 5$

D. $(x + 8)^2 + (y - 6)^2 = 5$

Q2

GEOMETRY AND TRIGONOMETRY

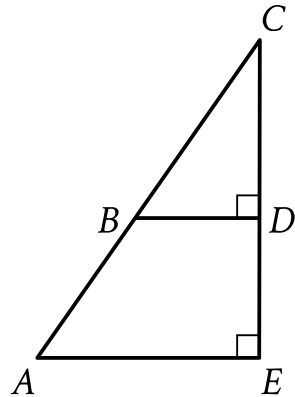
A right triangle has legs with lengths of 28 centimeters and 20 centimeters. What is the length of this triangle's hypotenuse, in centimeters?

A. $8\sqrt{6}$

B. $4\sqrt{74}$

C. 48

D. 1,184



Note: Figure not drawn to scale.

- Line segment BD is drawn from side CA to side CE to form triangle CBD .
- Angle CDB is a right angle.
- Angle CEA is a right angle.
- A note indicates the figure is not drawn to scale.

In the figure shown, triangle CAE is similar to triangle CBD . The measure of angle CBD is 57° , and $AE = 26BD$. What is the measure of angle CAE ?

- A. $26 \cdot 57^\circ$
- B. $26 + 57^\circ$
- C. 57°
- D. 26°

Q4

GEOMETRY AND TRIGONOMETRY

What is the length of one side of a square that has the same area as a circle with radius 2 ?

- A. 2
- B. $\sqrt{2\pi}$
- C. $2\sqrt{\pi}$
- D. 2π

Q5

GEOMETRY AND TRIGONOMETRY

What is the center of the circle in the xy -plane defined by the equation $x^2 + y^2 + 2x - 14y + 10 = 1$?

- A. -1, -7
- B. -1, 7
- C. 1, -7
- D. 1, 7

Q6

GEOMETRY AND TRIGONOMETRY

The length of a rectangle's diagonal is $5\sqrt{17}$, and the length of the rectangle's shorter side is 5. What is the length of the rectangle's longer side?

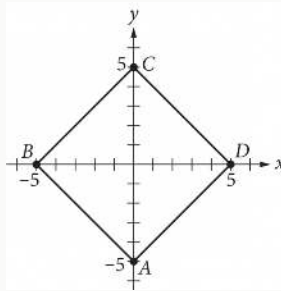
- A. $\sqrt{17}$
- B. 20
- C. $15\sqrt{2}$
- D. 400

Q7

GEOMETRY AND TRIGONOMETRY

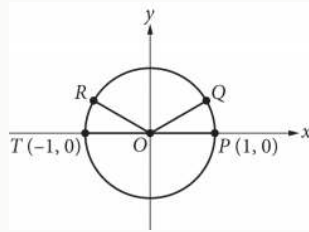
In triangle DEF , the measure of angle D is 47° and the measure of angle E is 97° . In triangle RST , the measure of angle R is 47° and the measure of angle S is 97° . Which of the following additional pieces of information is needed to determine whether triangle DEF is similar to triangle RST ?

- A. The measure of angle F
- B. The measure of angle T
- C. The measure of angle F and the measure of angle T
- D. No additional information is needed.



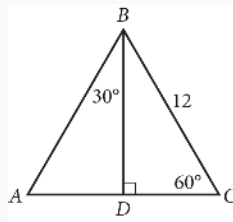
In the xy -plane shown, square ABCD has its diagonals on the x - and y -axes. What is the area, in square units, of the square?

- A. 20
- B. 25
- C. 50
- D. 100



In the xy -plane above, points P, Q, R, and T lie on the circle with center O. The degree measures of angles POQ and ROT are each 30° . What is the radian measure of angle QOR ?

- A. $\frac{5}{6} \pi$
- B. $\frac{3}{4} \pi$
- C. $\frac{2}{3} \pi$
- D. $\frac{1}{3} \pi$



In $\triangle ABC$ above, what is the length of $\overline{AD}$?

A. 4

B. 6

C.

$$6\sqrt{2}$$

D.

$$6\sqrt{3}$$